Naoki Kimura

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Selected Honors & Awards

- **Microsoft Research Asia Fellowship** (2020)
- **Google PhD Fellowship** (2019)
- **ACM CHI Best Paper Honorable Mention** (2019, 2018) – *Awarded to top 5% of submissions*
- **JST ACT-X Researcher** (2019 – 2022) – *Grant: Frontier of Mathematics and Information Science*
- **Toyota / Dwango AI Scholarship** (2018 – 2019)
- **JSPS Research Fellowship for Young Scientists (DC2)** (2021)

Education

The University of Tokyo | Tokyo, Japan *Ph.D. in Human-Computer Interaction* | 2019 – 2025 - Advisor: **Prof. Jun Rekimoto** - Graduate School of Interdisciplinary Information Studies

The University of Tokyo | Tokyo, Japan *M.S. in Applied Computer Science (Highest Distinction)* | 2013 – 2019 - Advisor: **Prof. Jun Rekimoto**

Professional Experience

LY Corporation Research (formerly Yahoo! JAPAN Research) | Tokyo, Japan *Chief Researcher* | Apr 2023 – Present - Leading advanced research projects in HCI and AI.

Google Research | Tokyo, Japan *Collaborative Researcher* | 2020 – 2021 - Advisor: **Dr. Shumin Zhai** - Research on mobile text entry and silent speech interfaces.

Microsoft Research Asia (MSRA) | Beijing, China (Remote) *Collaborative Researcher (PhD Fellow)* | 2020 - Advisor: **Dr. Frank Soong**

Georgia Institute of Technology | Atlanta, USA *Visiting Researcher* | 2019 – Present - Advisor: **Prof. Thad Starner** - Collaborating on wearable computing interfaces.

Preferred Networks (PFN) | Tokyo, Japan *Intern* | 2018 - Advisor: **Dr. Fabrice Matulic** - Contributed to the development of Chainer.

Selected Publications

[**N. Kimura**], "Rank-Residual Coding with a Shared Decoder: Lossless Low-Bit Communication for Ambiguous-Keyboard AAC," *IEEE SLT 2026*, 2026. **(Accepted)**

Naoki Kimura. "3-Key-Input: Exploring the Theoretical Minimum Keys for Text Entry." *ICASSP 2026*, 2026, pp. 19392-19396.

Kei Asano, **Naoki Kimura**, Jun Rekimoto. "HMDspeller: Fast and Hands-free Text Entry System for Head Mount Displays using Silent Spelling Recognition." *CHI 2023 Extended Abstracts*, 2023, pp. 1-4.

Naoki Kimura. "Self-Supervised Approach for Few-shot Hand Gesture Recognition." *UIST 2022 Adjunct Proceedings*, 2022, pp. 1-4.

Naoki Kimura, Tan Gemicioglu, Jonathan Womack, Richard Li, Yuhui Zhao, Abdelkareem Bedri, Zixiong Su, Alex Olwal, Jun Rekimoto, Thad Starner. "SilentSpeller: Towards mobile, hands-free, silent speech text entry using electropalatography." *CHI 2022*, 2022, pp. 1-19.

Naoki Kimura, Tan Gemicioglu, Jonathan Womack, Richard Li, Yuhui Zhao, Abdelkareem Bedri, Alex Olwal, Jun Rekimoto, Thad Starner. "Mobile, Hands-free, Silent Speech Texting Using SilentSpeller." *CHI 2021 Extended Abstracts*, 2021, pp. 1-5.

Naoki Kimura, Zixiong Su, Takaaki Saeki. "End-to-End Deep Learning Speech Recognition Model for Silent Speech Challenge." *INTERSPEECH 2020*, 2020, pp. 1025-1026.

Naoki Kimura, Michinari Kono, Jun Rekimoto. "SottoVoce: An Ultrasound Imaging-Based Silent Speech Interaction Using Deep Neural Networks." *CHI 2019*, 2019, pp. 1-11. **(Honorable Mention Award)**

Naoki Kimura, Jun Rekimoto. "ExtVision: Augmentation of Visual Experiences with Generation of Context Images for a Peripheral Vision Using Deep Neural Network." *CHI 2018*, 2018, pp. 1-10. **(Honorable Mention Award)**

Skills

- **Languages:** Japanese (Native), English (Professional Proficiency)
- **Technical:** Python, PyTorch, Chainer, C++, Unity, Signal Processing (EMG, Ultrasound), LLM Integration